

Course Code and Title: Applied Machine Learning Lab

Semester : Fall 2025

Instructor: Anoosha Khan | Sughra Kamran

Total Marks: 80

Deadline: 30TH November , 2025

CLOs		Description	Taxonomy Levels	PLO Levels
LLO3	Lab	Apply the knowledge of subject in the lab environment.	C3	1
LLO5	Lab	Build a product or system meeting the requirements.	P5	3

Project Statement — Applied Machine Learning

Instructions

- Work in groups of 2 members only.
- Select a dataset from Kaggle containing at least 35–40 features (columns).
- Using Python, apply a wide range of analytical and modeling techniques covered in your labs.
- Where specified, include both library-based implementations *and* from-scratch implementations.
- Perform a comparative analysis of each algorithm's performance in the final report: discuss results, similarities, differences, and reasons for performance variations.

Project Tasks

1. Exploratory Data Analysis (EDA)

- **Perform** detailed data exploration using descriptive statistics, visualization, feature distributions, correlation analyses, missing value handling, and outlier detection.
- **Display** insights effectively using visual tools such as plots and heatmaps.

2. Hypothesis Formulation, Cost Function & Gradient Descent

- **Formulate** a hypothesis related to the target variable.

- **Implement** a cost function and **apply** gradient descent (from scratch) to **demonstrate** understanding of parameter optimization.

3. Linear Regression

- **Implement** linear regression models (both from scratch and using a library).
- **Display** relationships between features and target variables; **interpret** coefficients and **evaluate** model fit.

4. Logistic Regression

- For classification datasets, **develop** logistic regression models (scratch + library).
- **Apply** thresholding and probability interpretation to **demonstrate** classification behavior.

5. Feature Regularization to Prevent Overfitting

- **Apply** regularization methods such as Ridge and Lasso regression to **manage** model complexity and avoid overfitting.
- **Demonstrate** the effect of regularization parameters on model performance.

6. Support Vector Machine (SVM)

- **Implement** an SVM classifier for your dataset.
- **Demonstrate** understanding of kernels, hyperplanes, and margins.
- **Evaluate** and **display** classification accuracy and model robustness.

7. Classifier Models: Naïve Bayes, K-Nearest Neighbors (KNN), Decision Trees, Random Forests

- **Apply** and **compare** these classifiers on your dataset.
- **Display** and **analyze** their performance; **demonstrate** understanding of their strengths and limitations.

8. Single-Layer and Multi-Layer Perceptron Models (Neural Networks)

- **Implement** a single-layer perceptron and **extend** it into a multi-layer perceptron (MLP).
- Incorporate Convolutional Neural Networks (CNNs) for comparison.
- **Demonstrate** how ANN and CNN models differ in learning complex data relationships and **display** comparative performance results.

9. Model Evaluation and Validation

- **Apply** appropriate validation techniques such as K-Fold Cross-Validation.
- **Demonstrate** understanding of evaluation metrics (e.g., accuracy, precision, recall, MSE, R^2).
- **Display** comparative results for each model using tables and graphs.

10. Unsupervised Learning: Clustering Analysis (Open-Ended Lab)

- **Apply** clustering algorithms such as K-Means and hierarchical clustering.
- **Display** clusters visually and interpret the discovered patterns.

11. Dimensionality Reduction – LDA & PCA

- **Apply** Principal Component Analysis (PCA) and Linear Discriminant Analysis (LDA).
- **Demonstrate** how dimensionality reduction impacts performance and visualization.
- **Display** results to manage data complexity effectively.

12. Additional Machine Learning Algorithms

- Incorporate two additional algorithms:
 - One related to hierarchical clustering (variant or deeper implementation).
 - One selected through research and applied to your dataset.
- **Compare** and **display** their performance against previously implemented models.

Report Requirements

- Provide a comprehensive report documenting all implemented methods and results.
- Include a detailed comparative analysis of all models and techniques: similarities, differences, and reasons for performance variations.
- Use tables, graphs, and visualizations where helpful to illustrate your comparisons.
- Include interpretation of results, key findings, and insights drawn from your analyses.

Submission Requirements

- Submit a Jupyter Notebook (.ipynb) containing all code and analysis.
- Submit a project report (PDF or Word) that covers methodology, comparisons, results, and discussion.